SETaP Week 5: From User Requirements to System Requirements

Let's assume you have extracted a set of user requirements from your notes. They might look like this:

"*Personal Tutors should be able to escalate issue to Student Monitoring Officer is:*

a. *Student has missed 3 consecutive PT meetings or*
b. *Student has missed 3 non-consecutive PT meetings without a valid reason."*

Task 1: Identify the concepts, actions, and elements of context.

"*Personal Tutors should be able to escalate issue to Student Monitoring Officer is:*

c. *Student has missed 3 consecutive PT meetings or*
d. *Student has missed 3 non-consecutive PT meetings without a valid reason."*

Task 2: Define each concept in terms of:

Let's use *issue* as an example:

A. Representation

An issue is defined by:

- *StudentID*: references the student the issue is raised for
- *Date*: the date when the issue is raised
- *PT* - the PT raising the issue {needs the definition of a PT}
- *Notes* - the description of the issue and any context known to the PT that they wish to share
- *Progress* - the stage this issue is at

*More attributes can be added to the list as more user requirements are processed.

B. Valid/Invalid representations

StudentID:

- Must be exactly 7 characters, all digits. First digit must not be zero.
- Must be a registered studentID.
- Must be a studentID associated with the PT associated with the issue

Date

- Must be a valid date in the format dd/mm/yy

- Must be automatically assigned to an issue as the current date

PT
- Must be a valid PT from a predefined list including all PTs in the School
- Must be automatically assigned to the issue as the name associated with the user ID of the current session

Notes
- This is an optional attribute in the form of a string of maximum 225 characters

Progress
- Must be one of the following values: New, Student contacted, PT notified of response, Closed, Reopened.

C. Associations

An issue is associated with a studentID. A studentID may be associated with multiple issues.

An issue is associated with a PT. A PT may be associated with multiple issues.
Equally, a studentID is associated with a PT. {address redundancy}

D. Affordances

An issue can be escalated by a PT.
{Needs the definition of escalation}

*The list of affordances might grow as we cover more user requirements.

Task 3: Define each action in terms of:

Let's use *escalate* as an example:

A. Breakdown (if needed)

Is this an atomic action or can it be broken down into further actions?

Based on the definition of an issue, we know the following:

1. An issue can be created by a PT provided that the conditions in the context above are met
2. An issue can be reopened by a PT

The question now becomes which of these two scenarios is an escalation.

Let's look closer at scenario 1. The missed meetings condition is met, this needs to be raised to the SMO. The decision is between:

1.1. The system automatically flags the studentID to the PT. The PT then chooses to create an issue, manually inputting the values for the attributes of the issue.

Is this action escalation or creation of an issue?

Or

1.2. The system automatically flags the studentID to the PT, creates a template issue with all but the Notes attributes filled in, and sends it to the SMO. The PT is notified of this, but they don't need to input any information at this stage.

Is this action escalation or creation of a template issue?

Or

1.3. The system automatically flags the studentID to the PT, requests the PT to create an issue and submit it to the SMO.

Is this an escalation or request for escalation sent to PT?

Each of these options will lead to a different design/implementation. This is the point when we make a decision.

Now, let's look closer at scenario 2 and the impact of the decision we made for scenario 1 will have on scenario 2.

2.1. If we go for option 1.1, the question becomes will the system create another issue for studentID when the missed meeting conditions are met at a later stage or will it change the status of the issue already raised to 'Reopened'? Are there any exceptions?

Is this an escalation, an automatic creation of another issue, or just a change in the status of the issue?

2.2. In case of option 1.2, is it the PT that adds more context to the notes and changes the status of the issue?

Is this an escalation or just an edit of the notes and a change in the status of the issue?

2.3 Lastly, in case of option 1.3, does the PT create another issue for studentID when the missed meeting conditions are met at a later stage or do they just change the status of the issue already raised to 'Reopened'?

Is this an escalation, a creation of another issue, or just a change in the status of the issue?

Once we have made a decision, that becomes our system requirement.

So, let's assume we define an escalation as Scenario 1.3. *The system automatically flags the studentID to the PT, requests the PT to create an issue and submit it to the SMO*.

B. Prerequisites

The question here is what needs to happen before this action can be achieved.

- The studentID must meet the missed meeting conditions, i.e. 3 consecutive meetings missed or 3 non-consecutive meetings missed without a reason.

*For this to be checked, we will need to specify additional system requirements:
- When and how is this condition checked?
- How do we track the reason for missing a meeting - is this an attribute for meeting?

C. Definition

If the missed meeting condition is met, the system will assign an 'escalation' tag to the student.

The system sends the PT receives a notification to say that studentID has been added to the 'escalation list', i.e. the list of students associated with the 'escalation' tag.

The system adds a form to the studentID that the PT has access to. The form allows the input of all attributes of an issue. The link to this form should be marked in red and associated directly with the student.

The system will check the validity of all PT inputs.

The system will allow the PT to submit the form once inputs are confirmed.

Once the form is submitted, the system will notify the SMO of the issue being raised for the student. This notification should add the studentID to the escalation list on the part of the SMO.

D. Valid/invalid scenarios

For every step in the definition of the action, define what can go wrong and how will the system respond. For example:

If the missed meeting condition is met, the system will assign an 'escalation' tag to the student.

- What happens if the studentID already has this tag assigned to it? Is this a possibility and how will the system respond?
- What happens if the studentID flagged is not registered?
- What happens if the missed meeting condition is not met? Does the system provide any feedback on that?
- etc.

E. Concepts affected

Student, PT, SMO, issue, meeting *possibly others depending on the definition of the action

F. Availability

Only the PT responsible for studentID can escalate an issue.

The SMO only has visibility over the inputted values for the attributes of the issue and no other notes on the PT's side.

***All statements defined as part of Tasks 2&3 are your System Requirements.**